Spanning Tree

At the end of this episode, I will be able to:

1. Spanning Tree

Learner Objective: *Learn the need for and basic functionality of Spanning Tree Protocol (STP) in modern networks*

Description: In this episode, you will learn about the need for spanning tree protocol in typical switched networks of today. You will learn about the basic functionality of this protocol and its different versions.

- Spanning Tree Protocol (STP) is a network protocol used in modern switched networks to prevent loops and ensure a loop-free topology.
- STP works by designating a single "root" bridge in the network, and then dynamically blocking redundant paths, ensuring that there is only one active path between any two network devices.
- This loop prevention mechanism is crucial in Ethernet networks where loops can lead to broadcast storms and network congestion.
- STP continuously monitors the network, adapting to changes in the topology and automatically reconfiguring the forwarding paths if a link failure or addition occurs.
- Multiple variations of STP, such as Rapid Spanning Tree Protocol (RSTP) and Multiple Spanning Tree Protocol (MSTP), have been developed to improve convergence times and support multiple VLANs in modern switched networks.

Additional Resources:

Spanning Tree Protocol
https://en.wikipedia.org/wiki/Spanning_Tree_Protocol